Traverse Closure Report: Example with an angle blunder at C

Project	Example with an angle blunder at C	Date	2026-10-23
Party chief	-	Software	Traverse Doctor v0.1.0
Start point	A: N 5,000.000, E 5,000.000	Start direction	S 49°19'13" E (A-B)

DOES NOT MEET REQUIREMENTS. Precision 1:315 (required 1:10,000). Likely angle blunder at station C. The angular misclosure of +1°00'02.0" is over the allowable 34". Moving it all onto the angle at C (measured 97°45'44", would be 96°45'42") leaves a linear misclosure of 0.035 ft (1:95,377). The next best station leaves 11.461 ft. Check the field book for that angle, for example a misread degree or a wrong backsight.
Warning: Least squares reference variance 85745.31 fails the chi-square test (bounds 0.07 to 3.12). The standard deviations are too optimistic or there is a blunder.

Summary

The 5-station loop has a perimeter of 3329.36 ft. The angular misclosure is +1°00'02.0" against an allowable of 34", so each angle was corrected by -0°12'00.4". After balancing, the linear misclosure is 10.554 ft toward N 22°40'39" E, a precision of 1:315 against the required 1:10,000.

The least squares adjustment has 3 redundant observations and a reference variance of 85745.3, which fails the chi-square test. Largest standardized residuals: Angle at C, Distance E-A, Angle at D and 7 more. Coordinates in this report come from the compass rule adjustment, and the area is 757,354.3 sq ft (17.3865 ac).

The traverse does not meet the requirements. Likely angle blunder at station C. The angular misclosure of +1°00'02.0" is over the allowable 34". Moving it all onto the angle at C (measured 97°45'44", would be 96°45'42") leaves a linear misclosure of 0.035 ft (1:95,377). The next best station leaves 11.461 ft. Check the field book for that angle, for example a misread degree or a wrong backsight.
Template summary (no Claude API key set).

Angular closure

Angles sum to 541°00'02". Interior angles of a 5-sided loop should sum to 540°00'00". Misclosure +1°00'02.0", allowable 3 x 5" x sqrt(5) = 34". Correction per angle -0°12'00.40".

Station	Angle right	Balanced	Course	Azimuth	Bearing	Distance (ft)
A	106°15'57"	106°03'56.6"	A-B	130°40'47.0"	S 49°19'13.0" E	641.400
B	113°23'03"	113°11'02.6"	B-C	63°51'49.6"	N 63°51'49.6" E	705.160
C	97°45'44"	97°33'43.6"	C-D	341°25'33.2"	N 18°34'26.8" W	658.260
D	120°19'45"	120°07'44.6"	D-E	281°33'17.8"	N 78°26'42.2" W	628.610
E	103°15'33"	103°03'32.6"	E-A	204°36'50.4"	S 24°36'50.4" W	695.930

Latitudes, departures and compass rule

Course	Latitude	Departure	Corr. lat	Corr. dep	Adj. distance	Adj. bearing
A-B	-418.084	486.415	-1.8761	-0.7839	642.031	S 49°08'52" E
B-C	310.628	633.057	-2.0626	-0.8618	703.479	N 63°59'01" E
C-D	623.973	-209.676	-1.9254	-0.8045	656.693	N 18°41'39" W
D-E	125.915	-615.870	-1.8387	-0.7683	628.997	N 78°37'23" W
E-A	-632.694	-289.857	-2.0356	-0.8506	698.135	S 24°36'28" W

Course	Latitude	Departure	Corr. lat	Corr. dep	Adj. distance	Adj. bearing
Sum	+9.7383	+4.0691	-9.7383	-4.0691	3,329.360	

Linear misclosure = sqrt(9.7383² + 4.0691²) = 10.5542 ft toward N 22°40'39" E. Precision = 3,329.360 / 10.5542 = 1:315.

Adjusted coordinates

Station	Compass N	Compass E	Transit N	Transit E	LSQ N	LSQ E
A	5,000.000	5,000.000	5,000.000	5,000.000	5,000.000	5,000.000
B	4,580.040	5,485.631	4,579.988	5,485.530	4,581.617	5,486.763
C	4,888.605	6,117.826	4,889.183	6,117.434	4,892.521	6,118.404
D	5,510.653	5,907.346	5,510.278	5,907.376	5,514.074	5,905.099
E	5,634.729	5,290.707	5,635.612	5,290.385	5,635.226	5,287.959

Area by coordinates (Compass rule): 757,354.3 sq ft (17.3865 ac). Compass 757,354.3, transit 757,423.8, least squares 757,367.9 sq ft.

Least squares adjustment

Weights from sigma angle 5" and sigma distance 0.01 ft + 3 ppm. A held fixed with the direction of A-B. Converged in 4 iterations. Redundancy 3, reference variance 85745.312 (sigma0 292.823). Chi-square test at 95%: 0.072 to 3.116, fails. Standardized residuals over 3.29 are flagged.

Observation	Observed	Residual	Std. residual	Redundancy	Flag
Angle at A	106°15'57"	+0°01'41.8"	+29.93	0.46	BLUNDER?
Angle at B	113°23'03"	-0°16'15.7"	-293.68	0.44	BLUNDER?
Angle at C	97°45'44"	-0°29'46.6"	-507.18	0.50	BLUNDER?
Angle at D	120°19'45"	-0°16'53.1"	-314.19	0.42	BLUNDER?
Angle at E	103°15'33"	+0°01'11.5"	+20.82	0.47	BLUNDER?
Distance A-B	641.400	+0.4586	+120.09	0.14	BLUNDER?
Distance B-C	705.160	-1.1495	-295.87	0.14	BLUNDER?
Distance C-D	658.260	-1.1241	-293.56	0.14	BLUNDER?
Distance D-E	628.610	+0.3096	+80.86	0.14	BLUNDER?
Distance E-A	695.930	+1.5168	+391.50	0.14	BLUNDER?

Standard error ellipses (scaled by the reference variance)

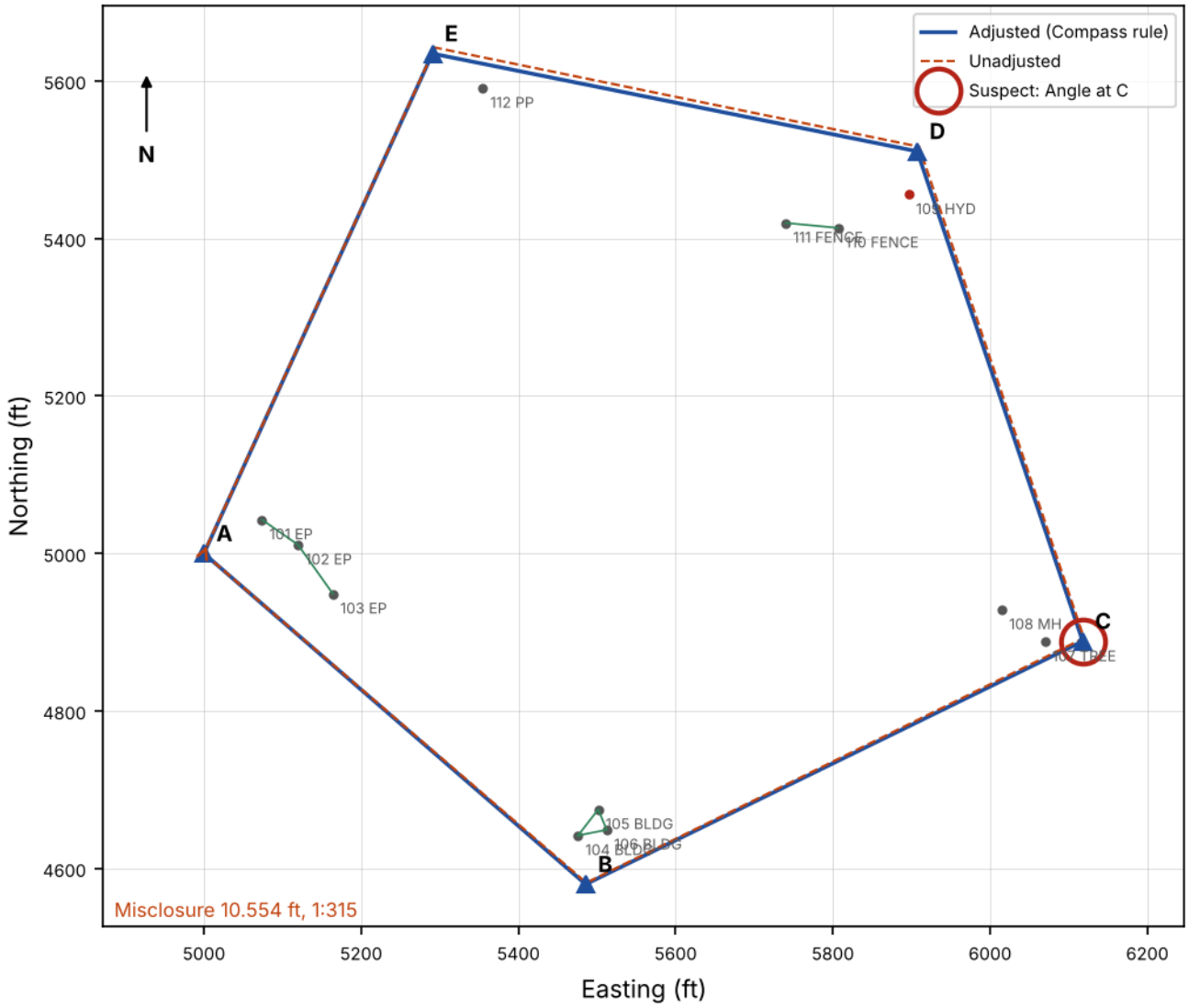
Station	Sigma N (ft)	Sigma E (ft)	Semi-major	Semi-minor	Major axis az.
A	0.0000	0.0000	0.0000	0.0000	fixed
B	1.8019	2.0964	2.7644	0.0009	130°40'47"
C	3.9258	3.0372	4.0368	2.8880	160°31'39"
D	3.8345	4.0419	4.8298	2.7774	131°59'47"
E	2.6208	3.7360	3.7417	2.6126	94°26'04"

Blunder search

Candidate	Fix	Misclosure left (ft)	Precision after
Angle at C	-1°00'02.0"	0.035	1:95,377
Angle at D	-1°00'02.0"	11.461	1:290
Angle at B	-1°00'02.0"	12.310	1:270

Candidate	Fix	Misclosure left (ft)	Precision after
Angle at E	-1°00'02.0"	19.462	1:171
Angle at A	-1°00'02.0"	19.640	1:170

Plan



The loop runs counterclockwise on the plot. If it looks mirrored compared with your sketch, the angles were turned the other way: switch between angles right and angles left.

Point list (from the compass rule coordinates)

Point	Northing	Easting	Code	Layer	Description
A	5,000.000	5,000.000	CP	V-CTRL	Mag nail in curb
B	4,580.040	5,485.631	IPS	V-PROP-MON	5/8 in rebar set
C	4,888.605	6,117.826	IPS	V-PROP-MON	5/8 in rebar set
D	5,510.653	5,907.346	CP	V-CTRL	PK nail in asphalt
E	5,634.729	5,290.707	IPF	V-PROP-MON	1/2 in pipe found
101	5,042.994	5,073.811	EP1 B	V-ROAD-EDGE	Edge of pavement
102	5,011.114	5,119.665	EP1	V-ROAD-EDGE	
103	4,948.763	5,164.087	EP1 E	V-ROAD-EDGE	
104	4,642.028	5,475.881	BLDG B	V-BLDG	Building corner
105	4,675.013	5,501.562	BLDG	V-BLDG	

Point	Northing	Easting	Code	Layer	Description
106	4,649.184	5,512.332	BLDG C	V-BLDG	
107	4,888.171	6,069.628	TREE	V-SITE-VEG	24 in oak
108	4,928.746	6,014.821	MH	V-SSWR-MH	Sanitary manhole
109	5,455.943	5,896.371	HYD	V-UNK-HYD	Fire hydrant
110	5,413.444	5,806.457	FENCE B	V-SITE-FENC	Chain link
111	5,419.716	5,740.043	FENCE	V-SITE-FENC	
112	5,590.966	5,354.850	PP	V-POWR-POLE	Power pole

Codes not in the library: HYD (closest: FH)

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